

Mutations in G protein beta subunits promote transformation and kinase inhibitor resistance

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Activating mutations in genes encoding G protein (G) subunits occur in 4-5% of all human cancers, but oncogenic alterations in G subunits have not been defined. Here we demonstrate that recurrent mutations in the G proteins GNB1 and GNB2 confer cytokine-independent growth and activate canonical G protein signaling. Multiple mutations in GNB1 affect the protein interface that binds G subunits as well as downstream effectors and disrupt G interactions with the G dimer. Different mutations in G proteins clustered partly on the basis of lineage; for example, all 11 GNB1 K57 mutations were in myeloid neoplasms, and seven of eight GNB1 I80 mutations were in B cell neoplasms. Expression of patient-derived GNB1 variants in Cdkn2a-deficient mouse bone marrow followed by transplantation resulted in either myeloid or B cell malignancies. In vivo treatment with the dual PI3K-mTOR inhibitor BEZ235 suppressed GNB1-induced signaling and markedly increased survival. In several human tumors, mutations in the gene encoding GNB1 co-occurred with oncogenic kinase alterations, including the BCR-ABL fusion protein, the V617F substitution in JAK2 and the V600K substitution in BRAF. Coexpression of patient-derived GNB1 variants with these mutant kinases resulted in inhibitor resistance in each context. Thus, GNB1 and GNB2 alterations confer transformed and resistance phenotypes across a range of human tumors and may be targetable with inhibitors of G protein signaling.